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Project Title:	Microgravity and the Biology of Neural Stem Cells		
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Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
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	The discovery and characterization of self-renewing, multipotent neural precursor (“stem”) cells has raised hopes for their possible therapeutic utility. Ordinarily, such cells are propagated as liquid cultures in stationary flasks, but a novel alternative system makes use of the Rotating Wall Vessel (RWV) bioreactor, a horizontally-rotated cylindrical culture vessel equipped with a gas-exchange membrane to optimize oxygenation. We have been testing the hypothesis that the behavior of neural precursors – both their propagation in vitro and their engraftment in vivo – is altered when they are cultured under simulated microgravity in RWV bioreactors known as High Aspect Ratio Vessels (HARVs). We have been studying murine cells derived from embryonic day 16 (E16) striatum / subventricular zone (SVZ) tissue dissociated and cultured in a chemically-defined, serum-free medium with epidermal growth factor (EGF). We have exploited a wide range of powerful in vitro and in vivo approaches to investigate possible microgravity-induced differences in		

Task Description:	cellular proliferation, differentiation, and function and analyze neural precursor cell (i) growth, development, and gene expression in rotary cell culture, and (ii) differentiation and function after transplantation into the mouse central nervous system. During the current budget year, we have focused on the in vivo work, inspired by our previous demonstration that EGF-propagated SVZ precursors can engraft, disseminate widely, and differentiate to become functional oligodendrocytes. When myelin-deficient shiverer (shi) mutant mice (genetically deficient in myelin basic protein [MBP]) are used as transplant hosts, there is a widespread incorporation of donor-derived oligodendrocytes, and they produce what appears to be biochemically and morphologically wild-type myelin (as previously reported in one of our NASA-sponsored publications [Mitome et al., Brain 124:2147-2161, 2001]). Importantly, we are aware that radiation-induced encephalopathy in humans appears to primarily affect central nervous system (CNS) myelin in white matter tracts. Since cellular proliferation within the SVZ is known to be reduced after external irradiation, CNS radiation injury might involve the failure of endogenous repair mechanisms that normally replace white matter oligodendrocytes. Therefore, this year we began to develop a murine radiation model in order to eventually test whether the transplantation of exogenous neural precursors might provide a novel approach to this problem by supplying the irradiated CNS with healthy new cells. Thus, this aspect of our research is now directed to risk 32, questions (h) and (i) of the Critical Path Roadmap. The objective of this research is to systematically compare the properties of neural stem cells grown in usual static culture with those of cells raised under the reduced hydrodynamic stress of microgravity, a condition that is known to dramatically alter the biology of proliferating cells. We will exploit a wide range of powerful in vitro and in vivo approaches to investigate possible microgravity-induced differences in cellular proliferation, differentiation, and function. The recent characterization of neural stem cells has aroused considerable hope that these cells might be potentially useful for repair of the human central nervous system in degenerative neurological diseases, recovery after trauma, and as delivery vehicles for gene therapy. Our experiments will investigate the possible utility of microgravity conditions for the proliferation and expansion of such cells by analyzing their multipotentiality, differentiation capacity, and functional activity in vitro and in vivo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are studying the biology of undifferentiated “stem” cells, including the regulation of their proliferation and differentiation. Basic research of this kind is important for eventually understanding normal and abnormal growth and development in animals (including humans) and for potentially exploiting such cells as a means of repair through cell replacement or as vehicles engineered to deliver exogenous genes or trophic factors.
Task Progress:	Radiation Studies. For radiation studies, we chose mice as brain-irradiated transplant hosts, because (a) a murine model of radiation leukoencephalopathy had been published by McBride and his associates (see Daigle et al., Cancer Res 66:8859-8865, 2001), (b) our precursor cells are murine derived, (c) behavioral testing in mice has been standardized, (d) immunohistochemical reagents for neuropathological studies are readily available, and (e) the power of mouse genetics should allow for future mechanistic investigations at the molecular level. Groups of C57BL/6J and C3H/HeJ mice were anesthetized and then irradiated to the CNS to prescribed total doses of 25, 40, 50, and 60 Gy. All mice developed the expected alopecia in the tailored field (as expected for a tissue with self-renewal potential), but their feeding and weight were unchanged (as expected if the oropharynx was appropriately shielded from therapy). We tested mice 4 and 6 months after radiation using a validated test battery for quantitating overlapping domains of motor behavior, including open field (for testing spontaneous exploratory activity and postural / gait abnormalities), cage wire hang (for testing grip strength and balance), and rotarod (for testing coordination and balance). We found no objective behavioral deficits in irradiated mice, although C57BL/6J mice showed a trend on open field testing, with diminished spontaneous activity at higher irradiation levels (41 + 33 at 40 Gy, 37 + 14 at 50 Gy, 18 + 5 at 60 Gy; # of photocell beam breaks/5 min test interval; not statistically significant by one-way ANOVA, $F = 1.43$, $p = 0.29$). We also found no objective pathological lesions at any dose on hematoxylin & eosin or on luxol fast blue stains of cut sections from sacrificed mice. To our knowledge, there have been no previous behavioral or neuropathological analyses of this type in mice with presumed radiation-induced CNS damage. MRI Studies. As part of our quest for developing non-invasive, longitudinal assays for monitoring the structure and function of CNS white matter, we began to work with scientists at the Center for Comparative Neuroimaging here at the University of Massachusetts on the utility of magnetic resonance imaging for detecting disorders of myelination in mice. We studied the effects of myelination and diffusion times (tdiff) on diffusion-tensor-imaging (DTI) contrast in wild type and shi mutant mice, as well as shi transplanted with wild type neural precursor cells. Among other findings, T2-weighted images showed obvious differences ($27 \pm 4\%$) between wild type and shi mice in the corpus callosum, whereas transplant-derived myelination was detectable on anisotropy maps but not on T2-weighted images. The spatial distribution of anisotropy changes was consistent with the distribution of donor-derived myelination in individual mouse brains as verified immunohistochemically. These studies serve as a critical first step for further developing more sensitive detection protocols for demyelinating diseases.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Abstracts for Journals and Proceedings	Nair G, Tanahashi Y, Low HP, Billings-Gagliardi S, Schwartz WJ, Duong TQ "Diffusion tensor imaging of myelin-deficient shiverer mice transplanted with neural precursor cells" Proc Int Soc Magn Reson Med xx , Jan-2005
Abstracts for Journals and Proceedings	Nair G, Low HP, Billings-Gagliardi S, Schwartz WJ, Duong TQ "DTI at long diffusion time reveals increased sensitivity to diffusion anisotropy" Proc Int Soc Magn Reson Med xx , Jan-2005
Articles in Peer-reviewed Journals	Nair G, Tanahashi Y, Low HP, Billings-Gagliardi S, Schwartz WJ, Duong TQ. "Myelination and long diffusion times alter diffusion-tensor-imaging contrast in myelin-deficient shiverer mice." NeuroImage. 2005 Oct;28(1):165–74. http://dx.doi.org/10.1016/j.neuroimage.2005.05.049 (Originally reported in 2005 as "in press as of September 2005.") , Oct-2005